Where should the character act, and when?

XR directing examples and demonstrated character controls, grouped by how the target and timing are specified.

◇ XR directing example ◈ Published direction or edit ◈ Related tracking or automatic control — A paper can appear at more than one coordinate.

RT Real-time demo or measured controller throughput — NO No real-time result reported for this example

Y What sets the timing?

Runtime condition triggers an action

“When I clap, stop.”

“When the door opens, turn toward it.” / “Look at B when they start speaking.”

“When I clap, look at the pinned marker.”

RT **CLoSD · ICLR 2025**
Automatic controller: detect sitting on the sofa, then wait two seconds before switching to standing.

RT **DartControl · ICLR 2025**
Automatically advance on waypoint arrival. The controller reports 240 frames/s for text-conditioned locomotion.

NO **MaskedMimic · ACM TOG 2024**
Automatic controller: switch from approaching the chair to sitting once the character is within range (§8.2).

[Point] “When I clap, look here.”

“Look at the cup. When B says ‘go,’ look here.” [Point] / “When the door opens, leave the table and come here.”

Action order or recorded sync

“Pause, then move faster.” / “Take two short steps, then stop.”

NO **Iterative Motion Editing · SIGGRAPH 2024**
Edit 3. Guard the face with the hands after the kick. The face is a body-relative target.

NO **CoMo: Controllable Motion Generation · ECCV 2024**
Add a left-hand raise at the end of an existing run (Fig. 6).

NO **S³: Head and Eye Animation · SIGGRAPH 2024**
Tag an upward glance in a recorded transcript.

NO **Move as You Say · CVPR 2024**
Generate a forward walk followed by a walk back (Fig. 1a).

“Look at the cup, then at the door.”

“Go to marker one, then marker two.”

NO **S³: Head and Eye Animation · SIGGRAPH 2024**
Tag a windshield object in an audio-aligned transcript to bias head and eye gaze. An aversion tag biases an upward glance.

[Point twice] “Look here, then over here.”

[Point] “Look here before looking at the cup.” / “Go to the table, then come here.”

Duration, rate or start time

“Hold still for four seconds.” / “Walk half as fast.”

NO **SimMotionEdit · CVPR 2025**
Speed up the action in a source clip (Fig. 1).

NO **STMC · CVPR Workshop 2024**
Set overlapping time intervals for raised arms across a walk-to-jump transition. Also combine walking backward with eating.

NO **Iterative Motion Editing · SIGGRAPH 2024**
Edit 2. Speed up the kick in the existing clip.

NO **FineMotion · ICCV 2025**
Extend the stand after a jumping jack by one second in a source clip (Fig. 8).

“Look at the cup for two seconds.”

“Reach the pinned target in two seconds.”

RT **ARDY · ACM TOG / SIGGRAPH 2026**
Click a floor-plane root waypoint and set its frame offset. Interpolate multiple clicks into a root path.

NO **MaskedMimic · ACM TOG 2024**
Reach a 3D right-hand target within 2 s. Reconstruct a cartwheel from head/hand tracking.

NO **OmniControl · ICLR 2024**
Constrain wrist positions at selected frames; guide the pelvis along a specified path.

NO **Head-EyeK · IEEE TVCG 2025**
Coordinate eyes and head from fixation targets and start times. §4.2 also evaluates a windowed variant for real-time use.

[Point] “Reach here in two seconds.” / [Trace] “Walk this path slowly.”

[Point] “From the table, walk here in three seconds.”

No explicit timing constraint

“Keep walking.”

NO **MotionFix · SIGGRAPH Asia 2024**
Lower a raised hand in an existing clip using text (Fig. 1).

NO **CoMo: Controllable Motion Generation · ECCV 2024**
Edit the existing clip to deepen the knee bend (Fig. 6).

NO **Iterative Motion Editing · SIGGRAPH 2024**
Edit 1. Raise the existing side kick. The edit supplies no external target or timing constraint.

“Move near the table.” / “Look at the cup.” / “Stand beside the blue chair.”

NO **Cen et al.: Motion in 3D Scenes · CVPR 2024**
From a 3D scan: walk to the table near the sofa; stand up from the table (Fig. 1).

NO **Move as You Say · CVPR 2024**
Use text and a 3D scene to generate a curved walk to a chair or a walk to the door (Fig. 1).

“Walk to the pinned mark.”

NO **Physical Character-Scene Interactions · SIGGRAPH 2023**
Supply a target object bounding box. Separate simulation tasks cover sitting, lying down and carrying a box.

RT **CLoSD · ICLR 2025**
Strike a supplied target using the hand or foot specified in the text prompt (§5.2).

[Point] “Move here.” / “Look here.” / [Trace] “Follow this arc.”

NO **MHC: Directable Human Motion · ECCV 2024**
Pose tracking: reconstruct body motion from a headset and controllers. A separate joystick demo sets velocity and heading.

RT **QuestEnvSim · SIGGRAPH 2023**
Track headset and controller poses for live sitting and getting up (§4.7).

[Point] “Come here from the chair.”

No external spatial target | Target described in language | Target supplied (ID or coordinates) | Target from XR input (point, trace or pose) | Described target + embodied XR input

X Source of the spatial target

Placement follows the first demonstrated control in each label. A runtime trigger takes precedence over a timer. Recorded transcript tags count as synchronization.

The selected studies leave the combined named-to-pointed gaze switch on a live cue unverified. Short paper titles link to sources.